

Lesson 7.3: Sensitivity analysis

Information dynamics

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Uncertainty revisited

- ▶ quantities are rarely known precisely (eg, profit)

	o	$\bar{o}$	EMV
E	-1	-1	-1.0
D	30	-10	10.0

- ▶ profit = revenue - cost
- ▶ ranges for revenues and costs (\$M):

	low	high
revenue	20	40
cost	10	20

Question

How confident to drill given uncertainty in values?

Scenario 1: decisions sensitive to inputs

worst/best cases (drilling, oil present):

	revenue	cost	profit
worst case	20	20	0
best case	40	10	30

worst case:

	o	$\bar{o}$	EMV
E	-1	-1	-1.0
D	0	-20	-10.0

best case:

	o	$\bar{o}$	EMV
E	-1	-1	-1.0
D	30	-10	10.0

► unsure about decision

Scenario 2: decision insensitivity

worst/best cases (drilling, oil present):

	low	high		revenue	cost	profit
revenue	35	50	worst case	35	15	20
cost	10	15	best case	50	10	40

worst case:

	o	$\bar{o}$	EMV
E	-1	-1	-1.0
D	20	-15	2.5

► confident in our decision